Отчёт по учебной практике. Сопровождение и обслуживание программного обеспечения компьютерных систем.

День 6

2026

1. Установка и настройка ОС Ubuntu server

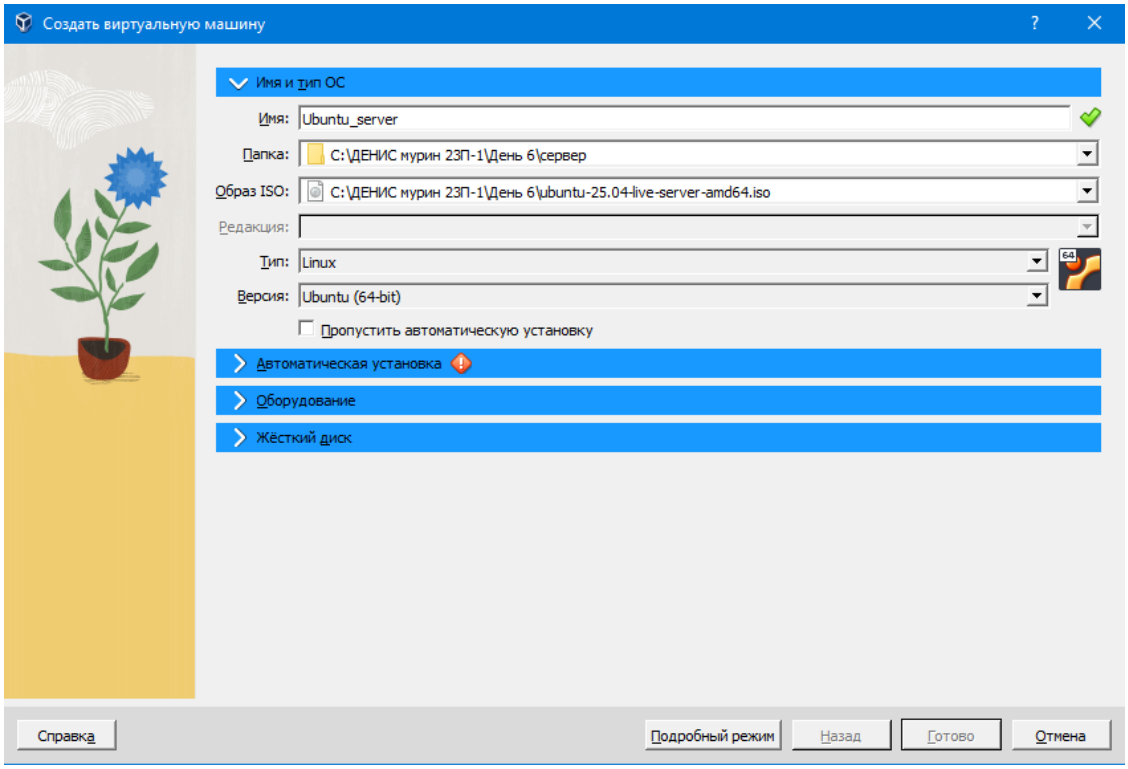

Рисунок 1 Создание ОС Ubuntu server

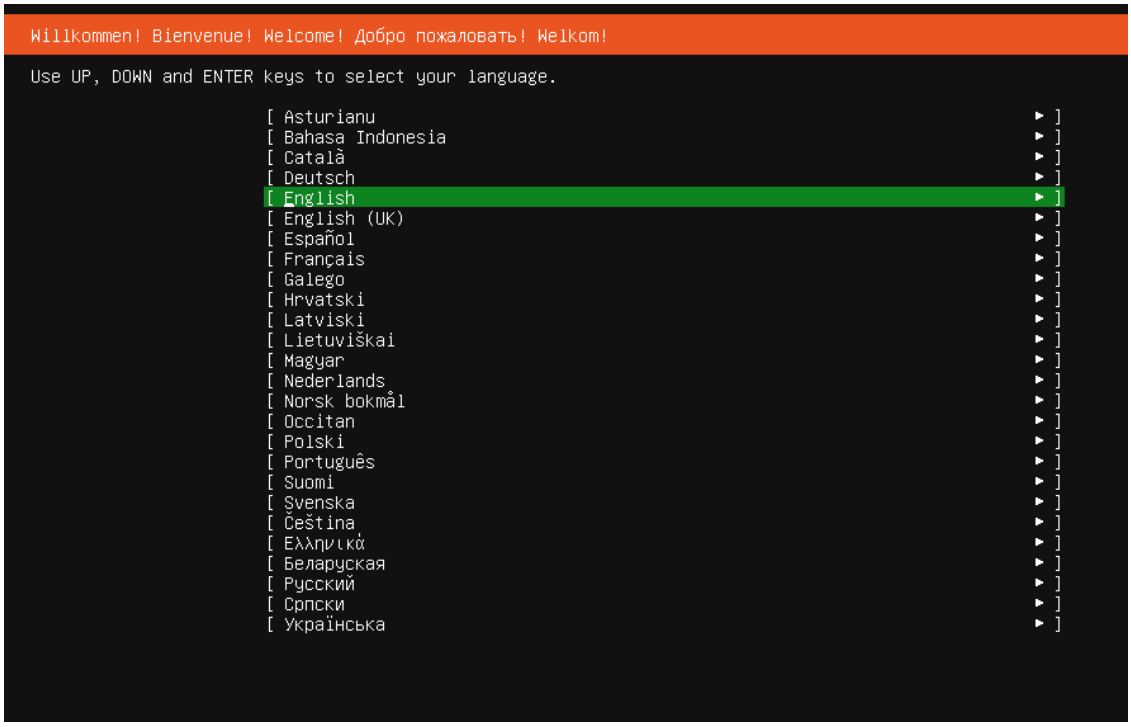

Рисунок 2 Выбор языка ОС Ubuntu server

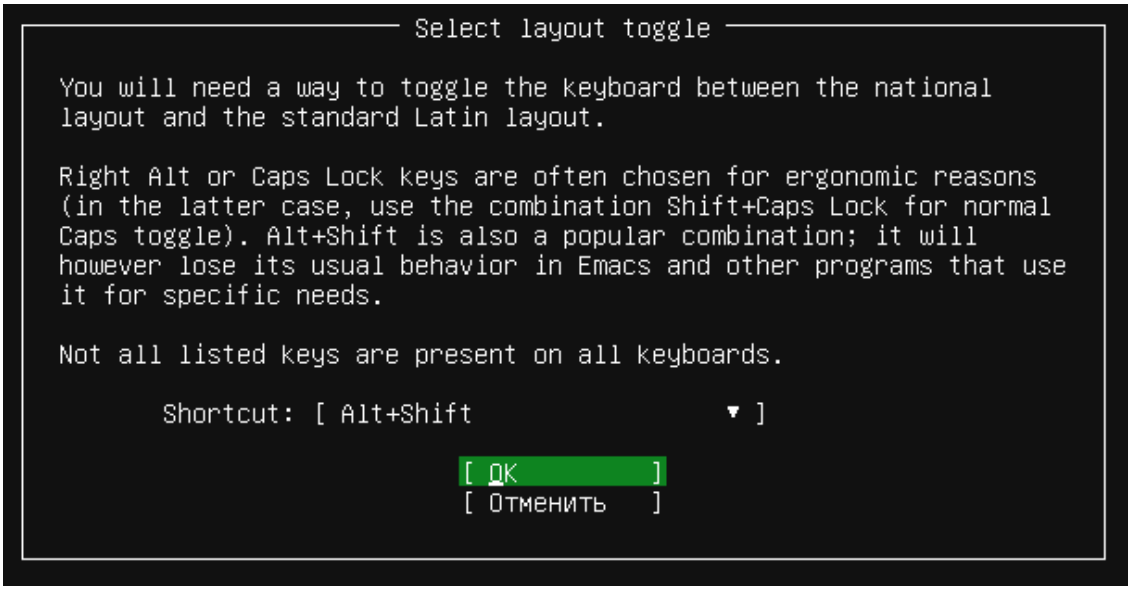

Рисунок 3 Выбор сочетая для переключения языка в ОС Ubuntu server

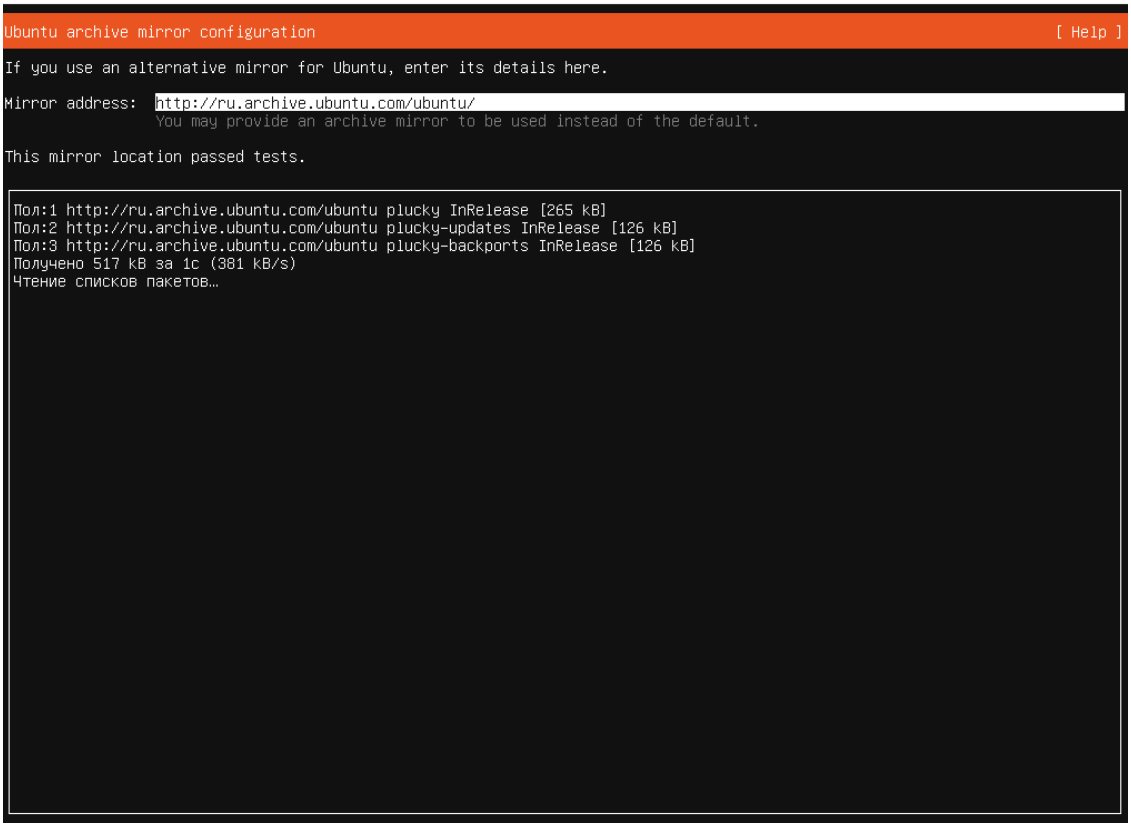

Рисунок 4 Зеркало

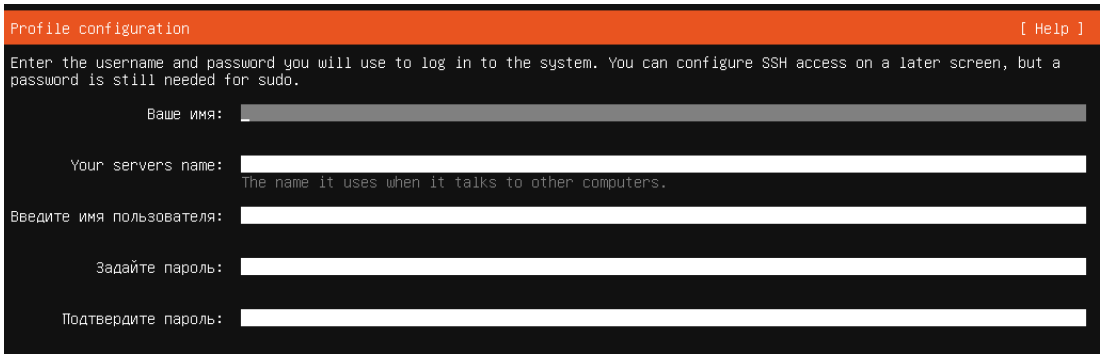

Рисунок 5 Окно создание пользователя в ОС Ubuntu server

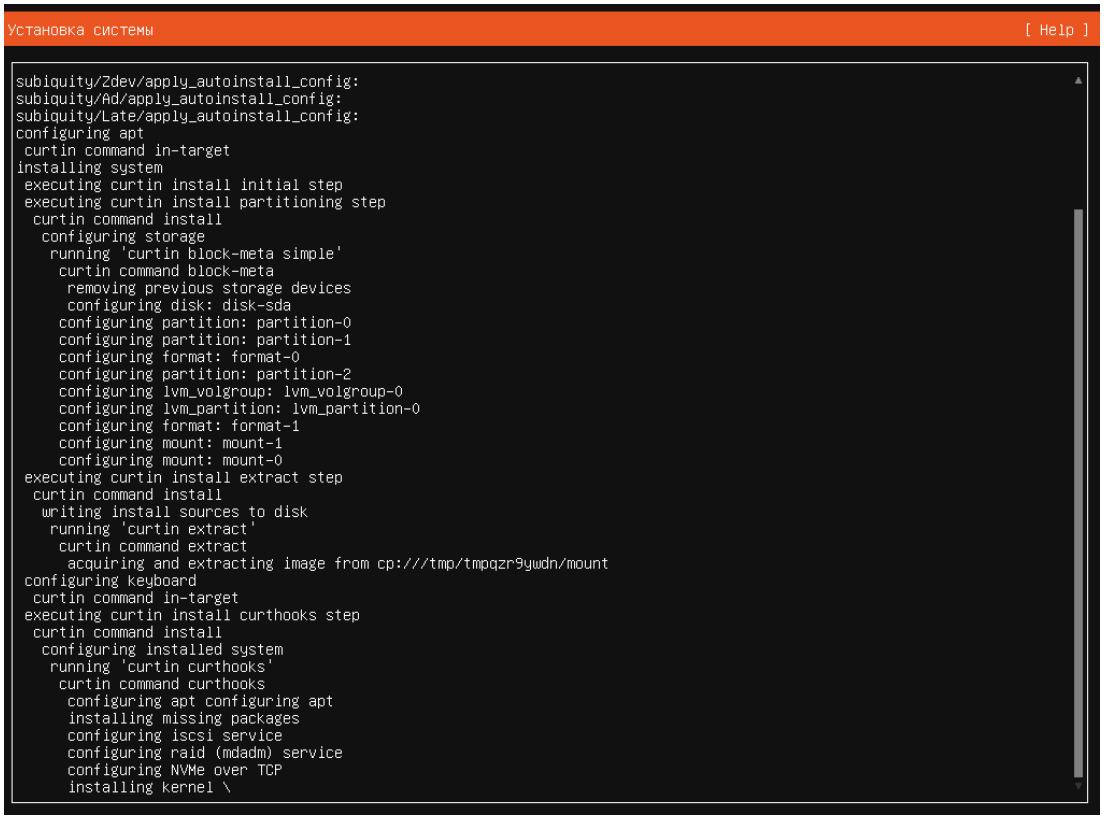

Рисунок 6 Установка ОС Ubuntu server

```
[   26.709985] cloud-init[851]: Your public key has been saved in /etc/ssh/ssh_host_rsa_key.pub
[   26.710085] cloud-init[851]: The key fingerprint is:
[   26.710123] cloud-init[851]: SHA256:LhYHkaKfa8PwToKfTmxNhIFACqIEeVqvYeAf6zSuw3w root@den
[   26.710155] cloud-init[851]: The key's randomart image is:
[   26.710199] cloud-init[851]: +---[RSA 3072]----+
[   26.710229] cloud-init[851]: |O=.   ..         |
[   26.710257] cloud-init[851]: |X oo. ..         |
[   26.710284] cloud-init[851]: |+=.o...          |
[   26.710312] cloud-init[851]: |..+o.  .         |
[   26.710340] cloud-init[851]: | ..++.. S        |
[   26.710367] cloud-init[851]: | ooBo  +         |
[   26.710395] cloud-init[851]: |+ O++.o .        |
[   26.710423] cloud-init[851]: | B E*. .         |
[   26.710451] cloud-init[851]: | oBo..           |
[   26.710483] cloud-init[851]: +----[SHA256]-----+
[   26.710542] cloud-init[851]: Generating public/private ecdsa key pair.
[   26.710583] cloud-init[851]: Your identification has been saved in /etc/ssh/ssh_host_ecdsa_key
[   26.710616] cloud-init[851]: Your public key has been saved in /etc/ssh/ssh_host_ecdsa_key.pub
[   26.710684] cloud-init[851]: The key fingerprint is:
[   26.710716] cloud-init[851]: SHA256:Fsx6fTjzFOVyiBvcnVWbrlATyImzoOrX3lTpv4r1Ue4 root@den
[   26.710747] cloud-init[851]: The key's randomart image is:
[   26.710780] cloud-init[851]: +---[ECDSA 256]---+
[   26.710817] cloud-init[851]: |          o o.. +|
[   26.710849] cloud-init[851]: |       + + * =.oo|
[   26.710878] cloud-init[851]: |      . = * =o=o |
[   26.716599] cloud-init[851]: |     . . + +o+o  |
[   26.722532] cloud-init[851]: |    . . S *+o  o |
[   26.726968] cloud-init[851]: |   .   o  o*. +  |
[   26.731435] cloud-init[851]: |  .   .  . o.o . |
[   26.734910] cloud-init[851]: |   . . .o o o o  |
[   26.738261] cloud-init[851]: |    . .. o ..+.E |
[   26.741943] cloud-init[851]: +----[SHA256]-----+
[   26.745391] cloud-init[851]: Generating public/private ed25519 key pair.
[   26.749260] cloud-init[851]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
[   26.753410] cloud-init[851]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
[   26.757248] cloud-init[851]: The key fingerprint is:
[   26.761739] cloud-init[851]: SHA256:Ll5GA90mDAReT/UADmUAwlT3jaGP1HZANiGqvXA18nM root@den
[   26.764978] cloud-init[851]: The key's randomart image is:
[   26.768942] cloud-init[851]: +--[ED25519 256]--+
[   26.770024] cloud-init[851]: | oo.+oB+%+o      |
[   26.775734] cloud-init[851]: |  .o = # B o     |
[   26.778510] cloud-init[851]: |    + O X = .    |
[   26.780783] cloud-init[851]: |   o = B +       |
[   26.785495] cloud-init[851]: |  o o + E        |
[   26.788700] cloud-init[851]: |   o . = .       |
[   26.791484] cloud-init[851]: |    . . +        |
[   26.795678] cloud-init[851]: |     . +         |
[   26.797956] cloud-init[851]: |      .          |
[   26.800763] cloud-init[851]: +----[SHA256]-----+
```

Рисунок 7 Загрузка файлов

```
den login: pikachu
Password:
Welcome to Ubuntu 25.04 (GNU/Linux 6.14.0-37-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Вт 28 апр 2026 06:05:27 UTC

  System load:  0.4               Processes:               120
  Usage of /:   30.7% of 9.75GB   Users logged in:         0
  Memory usage: 6%                IPv4 address for enp0s3: 10.0.2.15
  Swap usage:   0%

66 обновлений может быть применено немедленно.
Чтобы просмотреть дополнительные обновления выполните: apt list --upgradable



The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

pikachu@den:~$
```

Рисунок 8 Успешный вход в систему

2. Развертывание базы данных (Redis)

```
Установка:
  redis-server

Установка зависимостей:
  liblzf1  redis-tools

Предлагаемые пакеты:
  ruby-redis

Сводка:
  Обновление: 0, Установка: 3, Удаление: 0, Пропуск обновления: 0
  Объём загрузки: 1 384 kB
  Требуемое пространство: 7 584 kB / 6 659 MB доступно
```

Рисунок 9 Установка Redis на ОС Ubuntu server

```
pikachu@den:~$ sudo systemctl status redis
● redis-server.service - Advanced key-value store
     Loaded: loaded (/usr/lib/systemd/system/redis-server.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-04-28 06:36:17 UTC; 2min 39s ago
 Invocation: 78ecd57eeb6f49708ea694110315ce92
       Docs: http://redis.io/documentation,
             man:redis-server(1)
   Main PID: 1434 (redis-server)
     Status: "Ready to accept connections"
      Tasks: 5 (limit: 3980)
     Memory: 4.2M (peak: 4.6M)
        CPU: 269ms
     CGroup: /system.slice/redis-server.service
             └─1434 "/usr/bin/redis-server 127.0.0.1:6379"
```

Рисунок 10 Проверка работоспособности Redis

```
bind 192.168.1.100 10.0.0.1     # listens on two specific IPv4 addresses
bind 127.0.0.1 ::1              # listens on loopback IPv4 and IPv6
bind * -::*                     # like the default, all available interfaces
```

Рисунок 11 Просмотр bind

```
#
requirepass 1
```

Рисунок 12 Изменение requirepass

3. Установить клиент для работы с БД

```
апр 28 06:59:01 den systemd[1]: Starting redis-server.service - Advanced key-value store...
апр 28 06:59:01 den systemd[1]: Started redis-server.service - Advanced key-value store.
pikachu@den:~$ redis-cli -a "1"
Warning: Using a password with '-a' or '-u' option on the command line interface may not be safe.
127.0.0.1:6379> PING
PONG
127.0.0.1:6379>
```

Рисунок 13 Подключение к БД локально

```
login as: pikachu
pikachu@192.168.3.55's password:
welcome to Ubuntu 25.04 (GNU/Linux 6.14.0-37-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Вт 28 апр 2026 08:03:02 UTC

  System load:  0.01              Processes:               121
  Usage of /:   31.6% of 9.75GB   Users logged in:         1
  Memory usage: 8%                IPv4 address for enp0s3: 192.168.3.55
  Swap usage:   0%


0 обновлений может быть применено немедленно.


pikachu@den:~$
```

Рисунок 14 Подключение к БД удаленное

4. Импортировать данные в БД

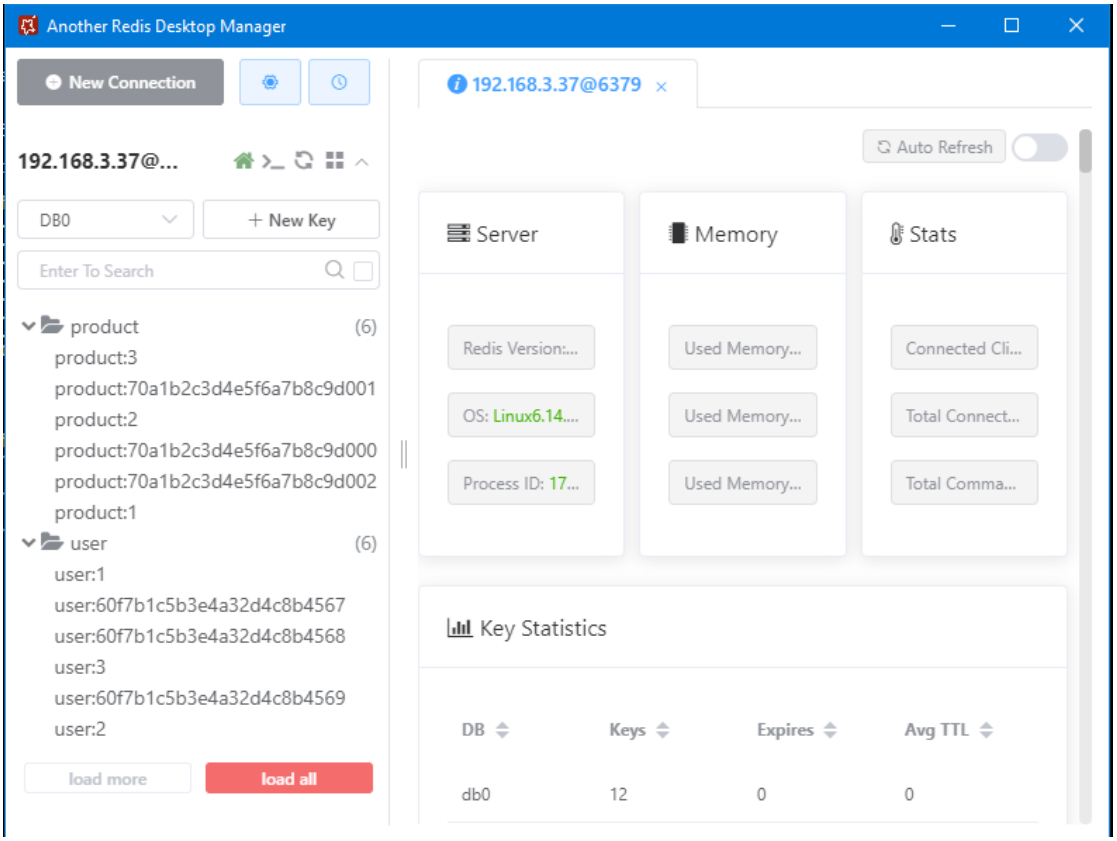

Рисунок 15 Приложении с импортом

5. Развернуть готовый API сервис на выбор с использованием Docker (Firecrawl)

нет

6. Развернуть Piper TTS + API Сервер на ubuntu server

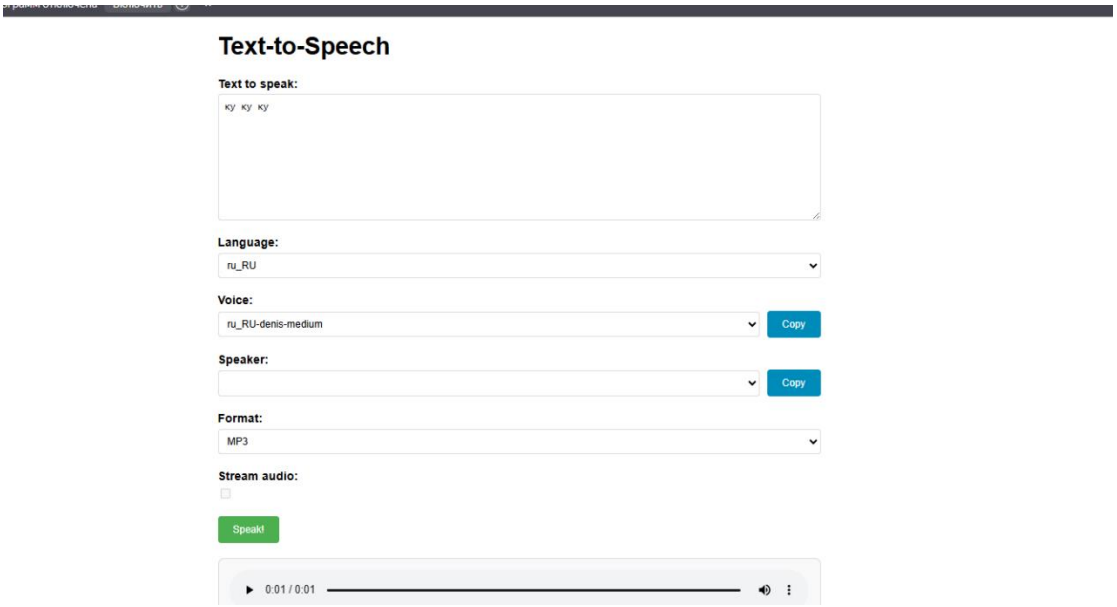

Рисунок 16 Выполненная задача